

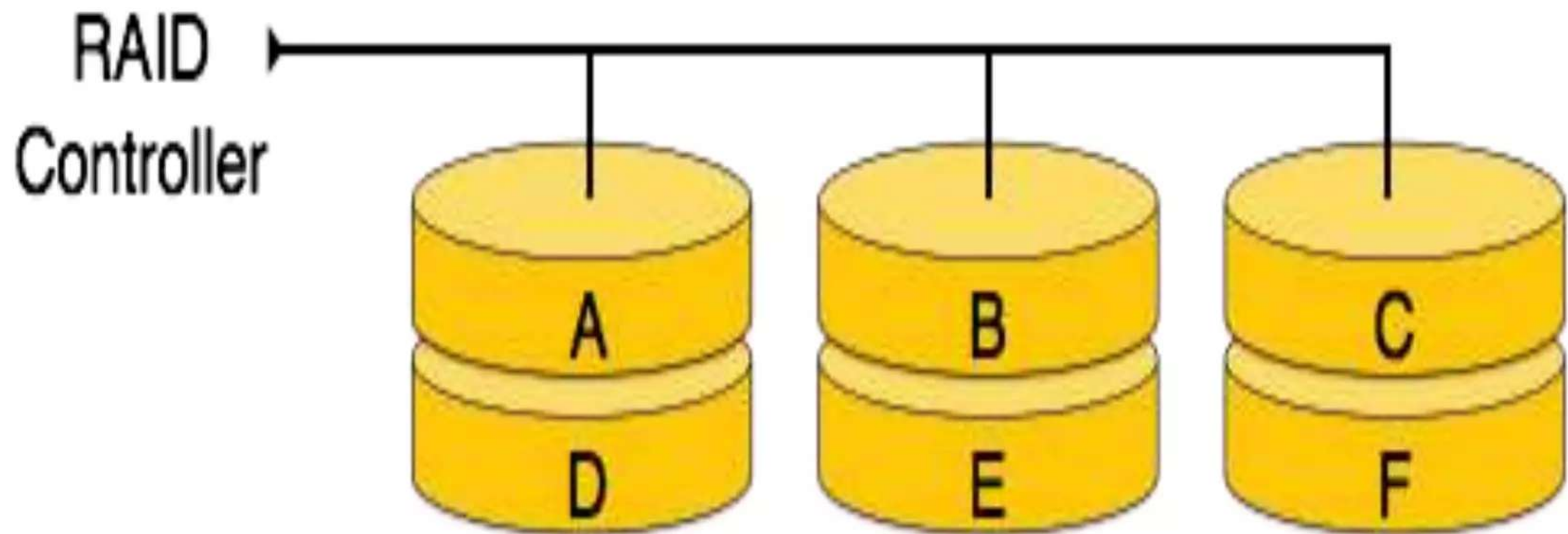
RAID

- Redundant Array of Independent Disks
- Multiple secondary disks are connected together to increase the performance, data redundancy or both.

RAID

- Need:
 - To increase the performance
 - Increased reliability
 - To give greater throughput
 - Data are restored

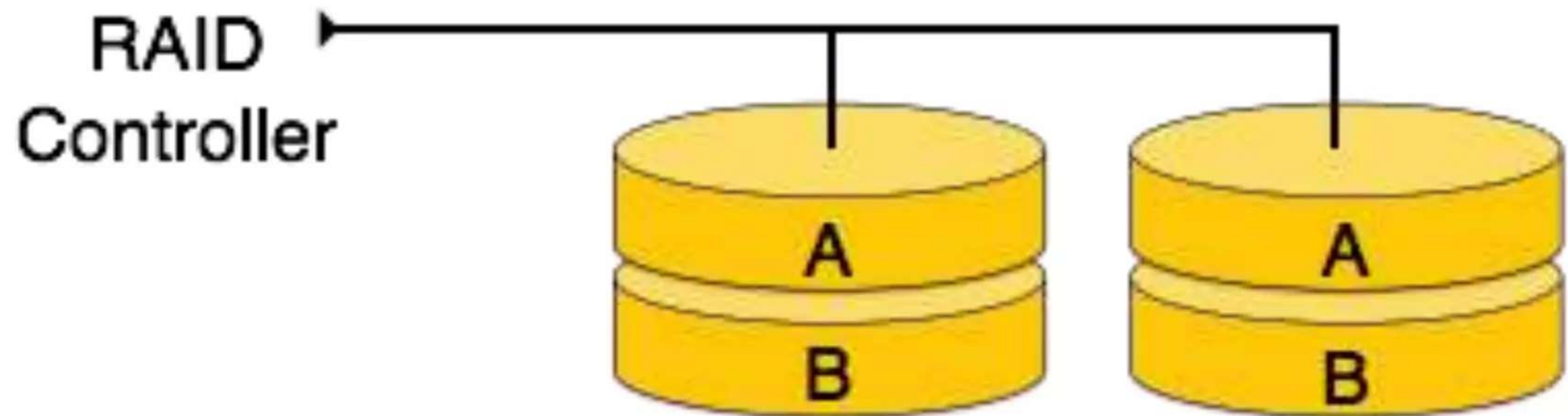
RAID – Level 0



RAID – Level 0

- Data is broken down into blocks and these blocks are stored across all the disks.
- Thus striped array of disks is implemented.
- There is no duplication of data in this level so once a block is lost then there is no way recover it.
- It has good performance.

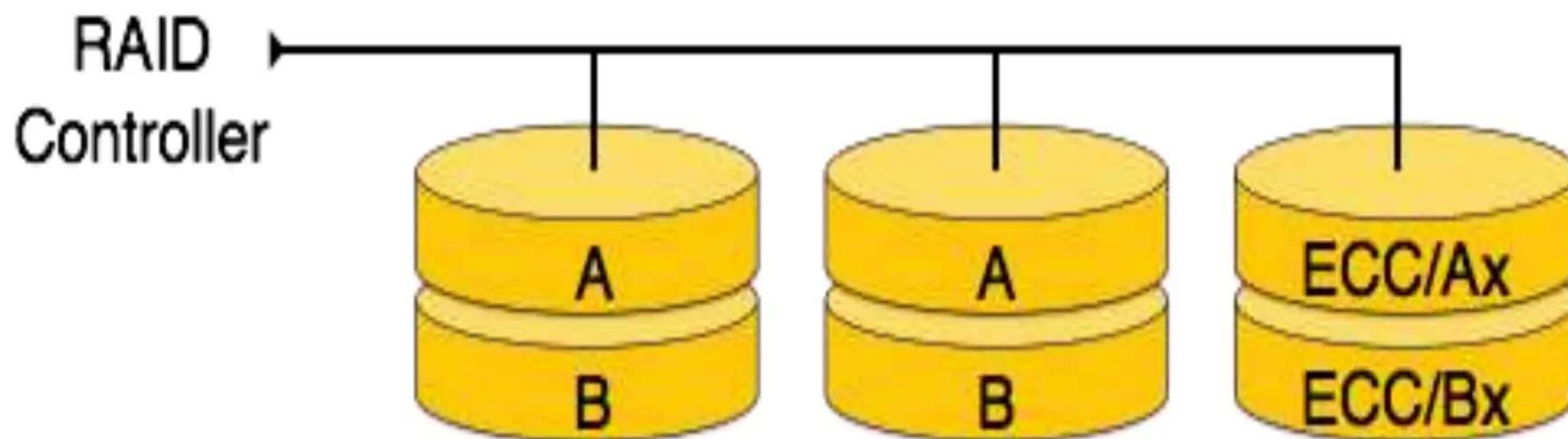
RAID – Level 1



RAID – Level 1

- uses mirroring techniques
- All data in the drive is duplicated to another drive.
- It provides 100% redundancy in case of a failure.
- Advantage: Fault Tolerance

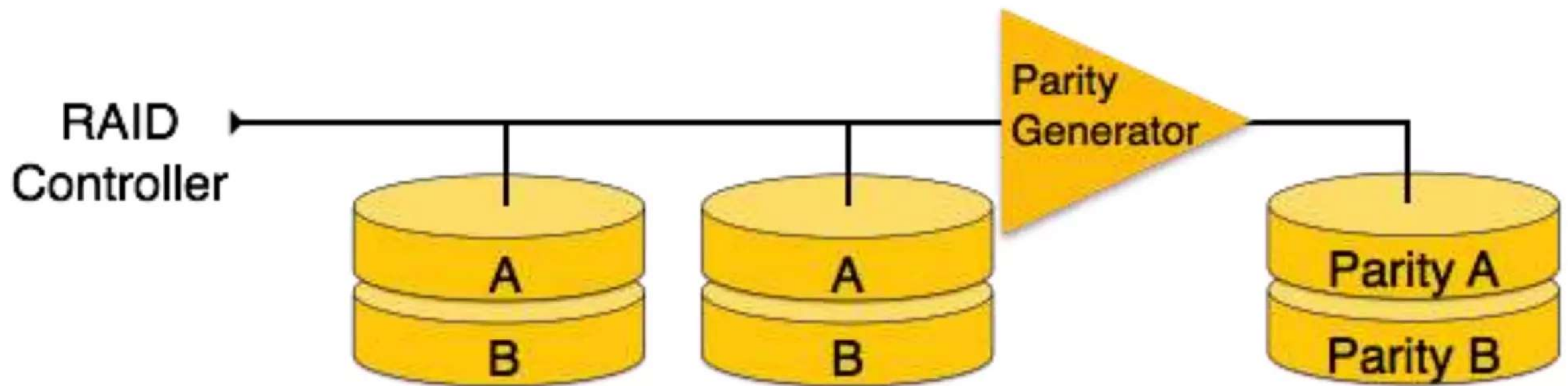
RAID – Level 2



RAID – Level 2

- Use of mirroring as well as stores Error Correcting codes for its data striped on different disks.
- Each data bit in a word is recorded on a separate disk and ECC codes of the data words are stored on a different set disks.
- Due to its complex structure and high cost, RAID 2 is not commercially available.

RAID – Level 3



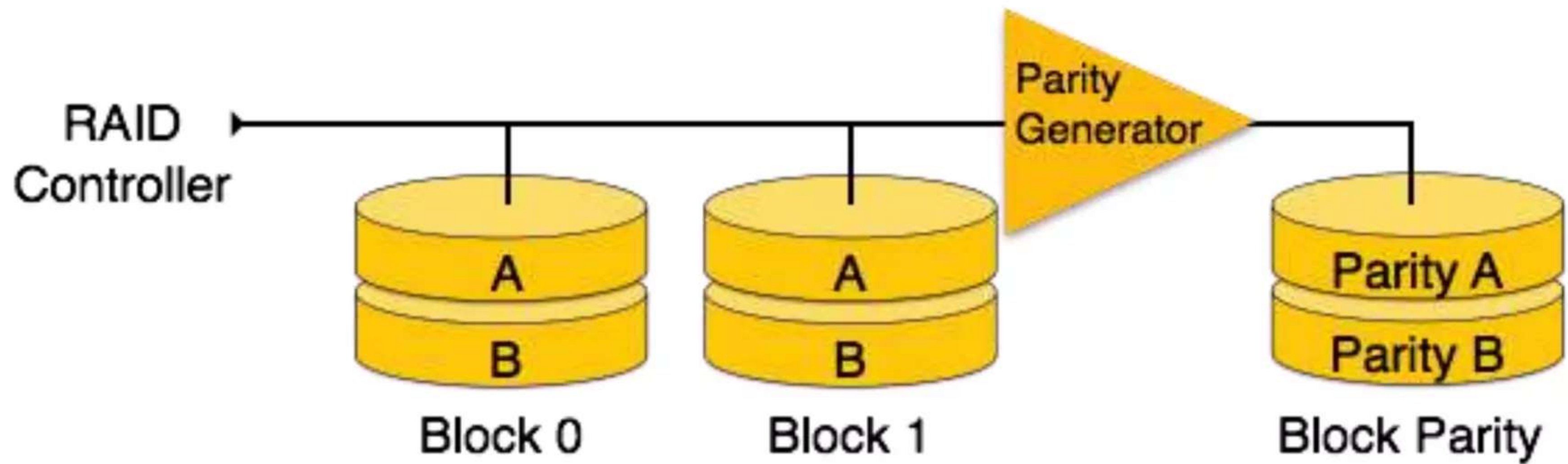
RAID – Level 3

- It consists of byte level stripping with dedicated parity. In this level, the parity information is stored for each disk section and written to dedicated parity drive.
- Parity is a technique that checks whether data has been lost or written over when it is moved from one place in storage to another.

RAID – Level 3

- In the case of disk failure, the parity disk is accessed and data is reconstructed from the remaining devices.
- Once the failed disk is replaced, the missing data can be restored on the new disk.

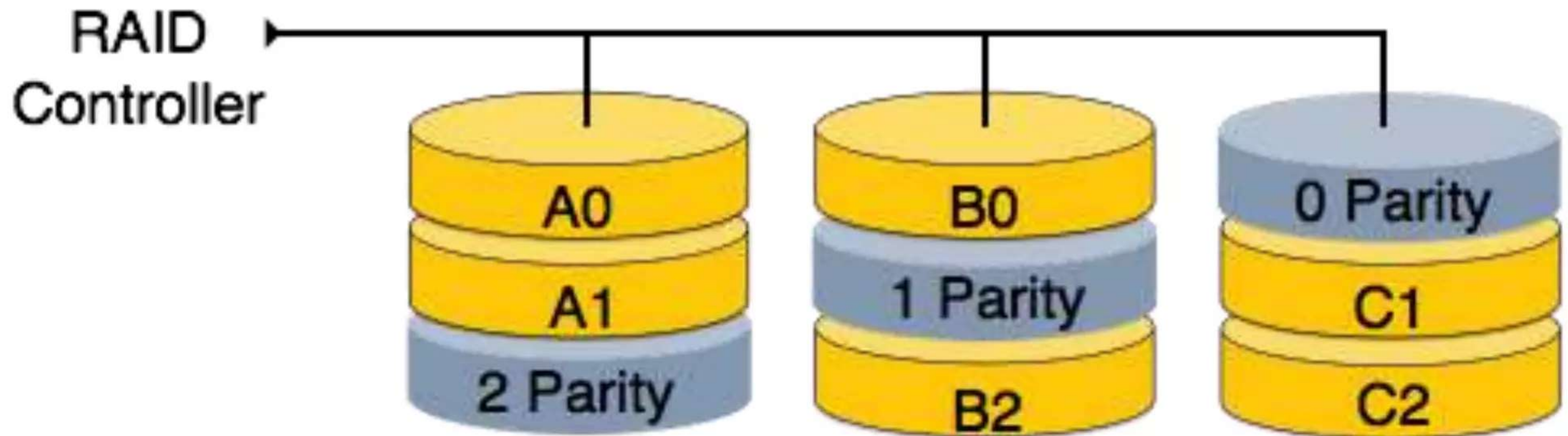
RAID – Level 4



RAID – Level 4

- It consists of block level stripping with a parity disk.

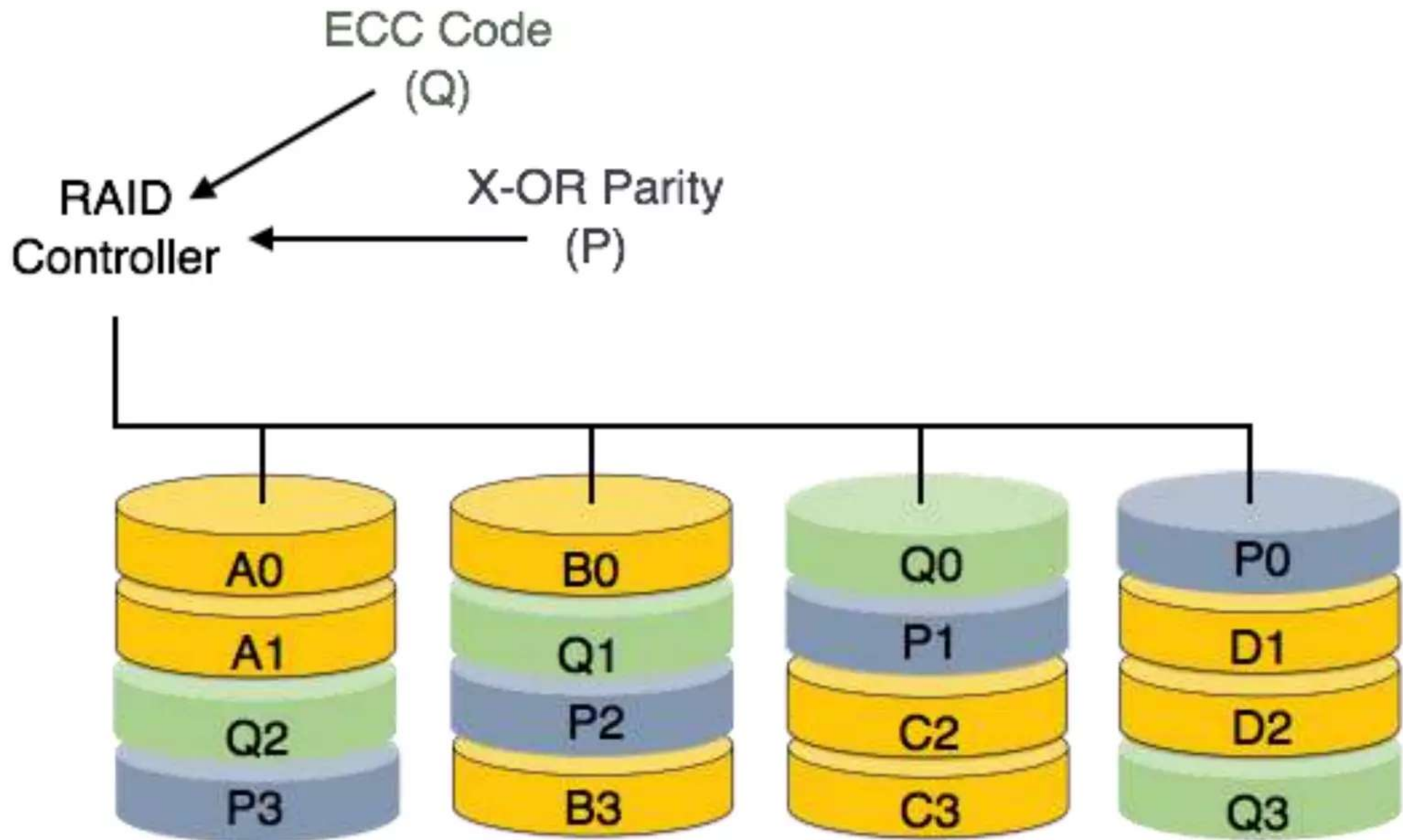
RAID – Level 5



RAID – Level 5

- RAID 5 writes whole data blocks onto different disks, but the **parity bits** generated for data **block stripe are distributed** among all the data disks rather than storing them on a different dedicated disk.

RAID – Level 6



RAID – Level 6

- RAID 6 is an extension of Level 5.
- In this level, **two independent parities** are generated and stored in distributed fashion among multiple disks.
- Two parities provide additional fault tolerance.
- This level requires at least four disk drives to implement RAID.

- The factors to be taken into account in choosing a RAID level are:
 - Performance requirements in terms of number of I/O operation.
 - Performance when a disk has failed.
 - Performance during rebuild.